# Blessed G/Michael Catholic Secondary School

## Semester 1 Chemistry Mid1 Examination for Grade9

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

**Name:** _____ **Grade & Sec:** Grade9 _____ **R.No:** _____

**I. Write 'True' if the statement is correct and 'False' if it is incorrect.**

____ **1.** Chemistry does not study the energy changes associated with chemical reactions.

____ **2.** Analytical chemistry is only concerned with producing chemicals, not analyzing them.

____ **3.** Industrial processes such as brewing, soap making, and dye production rely on chemical principles.

____ **4.** All measurements in chemistry must include a number and a unit.

**II. Choose the best answer among the given alternatives.**

____ **5.** The primary focus of chemistry is the study of:

**A.** Living organisms and their functions
**B.** The motion and forces affecting objects
**C.** The composition, structure, properties, and changes of matter
**D.** The history and evolution of the universe

____ **6.** Which of the following best describes the "matter" that chemists study?

**A.** Only the substances that can be seen and touched
**B.** Any substance that has mass and takes up space
**C.** Only the elements found on the Periodic Table
**D.** Only complex compounds, such as plastics and medicines

____ **7.** A chemist who is studying the rate at which a chemical reaction occurs is working in the branch of chemistry known as:

**A.** Analytical chemistry **B.** Inorganic chemistry
**C.** Physical chemistry **D.** Bio chemistry

____ **8.** The process of observing a substance and determining its components and their relative amounts is primarily the focus of which branch of chemistry?

**A.** Physical chemistry **B.** Analytical chemistry
**C.** Organic chemistry **D.** Bio chemistry

____ **9.** A researcher is studying a newly discovered compound containing only carbon and hydrogen atoms. This falls under the branch of chemistry called:

**A.** Inorganic chemistry **B.** Biochemistry
**C.** Organic chemistry **D.** Analytical chemistry

____ **10.** In industrial production, chemistry plays a critical role in converting raw materials into useful products such as:

**A.** Solar panels, wind turbines, and hydropower stations
**B.** Plastics, medicines, and fertilizers
**C.** Computer software, mobile apps, and video games
**D.** Books, newspapers, and magazines

____ **11.** In agriculture, chemistry is primarily used to:

**A.** Create new types of soil
**B.** Develop fertilizers and pesticides that increase crop yield
**C.** Build farming machinery
**D.** Control weather conditions

____ **12.** Petrochemical industries rely on chemistry mainly to:

**A.** Grow crude oil naturally
**B.** Refine crude oil into fuels and useful chemical products
**C.** Remove all carbon from fuels
**D.** Convert metals into plastics

____ **13.** Which of the following best defines the chemical industry?

**A.** The industry that produces machines and mechanical tools
**B.** The industry that extracts minerals from the earth
**C.** The industry that converts raw materials into useful products through chemical processes
**D.** The industry that provides banking and financial services

____ **14.** Which of the following is NOT a major sector of the chemical industry?

**A.** Petrochemical industry **B.** Pharmaceutical industry

**C.** Textile chemical processing **D.** Automobile repair industry

____ **15.** Which of the following is an example of a chemical product?

**A.** Crude oil directly from the ground
**B.** Water from a river
**C.** Plastic made from polymerization of ethene
**D.** Wood cut from a tree

____ **16.** The composition of matter refers to:

**A.** The physical appearance of matter
**B.** The types and proportions of atoms or elements that make up a substance
**C.** The amount of energy matter contains
**D.** The temperature of matter

____ **17.** Which of the following is not a basic physical quantity?

**A.** Mass **B.** Length **C.** Temperature **D.** Concentration

____ **18.** Temperature in the SI system is measured in:

**A.** Celsius (°C) **B.** Fahrenheit (°F)

**C.** Kelvin (K) **D.** Joule (J)

____ **19.** Which of the following quantities measures the space occupied by matter?

**A.** Density **B.** Volume **C.** Mass **D.** Pressure

____ **20.** Which of the following is a derived physical quantity in chemistry?

**A.** Time **B.** Mass

**C.** Density **D.** Amount of substance

____ **21.** A chemist measures 2.5 mol of oxygen gas. The term mole refers to:

**A.** The mass of oxygen gas **B.** The number of particles present

**C.** The temperature of oxygen **D.** The pressure of the gas

____ **22.** Temperature is a measure of:

**A.** The total energy of a system
**B.** The average kinetic energy of particles in a substance
**C.** The potential energy of atoms
**D.** The size of molecules

____ **23.** Which statement about the Kelvin scale is true?

**A.** It is based on the freezing and boiling points of water
**B.** It has negative temperature values
**C.** It begins at absolute zero where all molecular motion stops
**D.** It is used only in meteorology

____ **24.** Convert 0 °C to Fahrenheit.

**A.** 32 °F **B.** 0 °F **C.** 100 °F **D.** -32 °F

____ **25.** If a sample of gas is heated from 27 °C to 327 °C, by how many Kelvin does the temperature increase?

**A.** 100 K **B.** 200 K **C.** 300 K **D.** 600 K

____ **26.** At what temperature are Celsius and Fahrenheit numerical values equal?

**A.** 0 °C **B.** -32 °C **C.** -40 °C **D.** 40 °C